Day 10: Subnetting Test and Introduction to VLSM

Level 1, Month 1, Networking foundations

What I learned

VLSM (Variable Length Subnet Mask) means each subnet gets the mask that fits its size, so addresses are not wasted.

Steps:

1. Sort the networks from the largest to the smallest.
2. For each one, find the smallest `bits` such that `2^bits - 2` is at least the number of hosts. Check that the next smaller power is not enough.
3. Mask = `32 - bits`. Block size = `2^bits`.
4. Allocate the blocks one after another. Each block starts right after the previous broadcast.
5. Checks: the network address is a multiple of the block size, and the blocks do not overlap.

Test: 3 problems in a row with zero mistakes

Problem	Reseau	Premiere	Broadcast	Derniere	Ho
172.25.99.140/22	172.25.96.0	172.25.96.1	172.25.99.255	172.25.99.254	10
192.168.7.200/26	192.168.7.192	192.168.7.193	192.168.7.255	192.168.7.254	62
10.100.50.19/29	10.100.50.16	10.100.50.17	10.100.50.23	10.100.50.22	6

All three were correct. The change that made the difference: I wrote the table line for each problem (for example `100 > 99 > 96`).

VLSM exercise: 192.168.20.0/24 with 100, 50 and 10 hosts

Hosts needed	Bits	Mask	Block	Network	First	Broadcast	Last
100	7	/25	128	192.168.20.0	.1	.127	.126
50	6	/26	64	192.168.20.128	.129	.191	.190

Hosts needed	Bits	Mask	Block	Network	First	Broadcast	Last
10	4	/28	16	192.168.20.192	.193	.207	.206

Free space left for later: 192.168.20.208 to 192.168.20.255.

Error log

Mistake	Correct answer	Why
Broadcast of the second network: .189	.191	I added 62 (valid hosts) instead of 64 (block size). End of block = start + block size - 1.
Third network started at .190	.192	The start of the third block must follow the previous broadcast, and it must be a multiple of the block size (16).

Main lessons

- Move in the table by the block size, never by the number of hosts.
- A network address must be a multiple of its block size. This check catches a wrong start.
- Slowing down works: when I wrote every step, the three test problems were perfect.
- Overwriting digits makes my page hard to read. One step per line, cross out once and rewrite.

Anki

New cards in `anki-cards-10.txt`.

Next session

Another VLSM exercise as the review problem, then the OSI model (the 7 layers).